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A coupled climate model simulation of the Last Glacial Maximum, Part 1: transient multi-decadal response

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Abstract A coupled atmosphere–ocean–sea ice–land surface climate system model developed at the Canadian Centre for Climate Modelling and Analysis (CCCma) is used to investigate the response to glacial ice-sheet topography and decreased CO₂ representative of the last glacial maximum (LGM, roughly 21,000 years before present). Imposing these glacial boundary conditions leads to a strong and rapid climate adjustment, first in the atmosphere, and then in the ocean. We describe the model and boundary conditions and analyze the initial transient response. In a subsequent paper we analyze the near-equilibrium response. During the transient phase, surface air temperature rapidly decreases with time. The rapid cooling over the Laurentide and Fennoscandian ice sheets introduces a strong inter-hemispheric asymmetry. Sea surface temperature also decreases and after 80 years of integration, ocean surface temperature has decreased by 3 °C and global surface air temperature by 5 °C. The expected overall cooling is contrasted by marked regions of unexpected localized ocean surface warming and the retreat of sea ice in the northern North Atlantic and in the high latitude Southern Ocean. This strong transient behavior, associated with more vigorous oceanic convection at these high latitudes, represents the localization of the initial response in areas remote from the direct forcing change. The strength of the North Atlantic (Southern Ocean) overturning stream function, associated with the formation of North Atlantic Deep Water (Antarctic Bottom Water), increases from 12 to 36 Sv (2 to 26 Sv). There is a marked increase in the Antarctic Circumpolar Current transport as well (from 80 to 140 Sv), even though the wind stress decreases over the Southern Ocean, presumably as a consequence of the enhanced bottom pressure torque due to an increase in

the density and the formation of Antarctic Bottom Water. Both precipitation and evaporation decrease on average with time and the change in the hydrological cycle modifies sea surface salinity (SSS). SSS increases in the Arctic Ocean due to a decrease in river runoff while it decreases in the North Atlantic due to an increase in river runoff and an increase in precipitation minus evaporation. The initial (and near equilibrium) climate response to LGM forcing in the fully coupled atmosphere–ocean climate model is quite different from that in the associated atmosphere-mixed layer ocean model due to very strong oceanic feedbacks. This is especially evident where oceanic convection is active. The anomalously warm surface temperature, especially in the Southern Ocean, and the marked increase in the overturning circulation are transient features of the response to imposed LGM boundary conditions which moderate or reverse as the system approaches equilibrium (as described in the companion paper). Other differences with the mixed layer simulation include a considerably colder tropical Pacific and the development of a La Niña-like response both of which are absent in the mixed layer simulations.

1 Introduction

The Last Glacial Maximum (LGM) at about 21 kaBP (thousand years before present) provides a dramatic image of the non-constancy of the Earth's climate. Considerable effort has been devoted to the reconstruction of LGM climatic conditions based on geological proxy data (e.g., CLIMAP 1976, 1981). Many lines of geological evidence provide a broad picture of the LGM climate, but each has limitations in revealing the features and factors involved in such a severe event. As an adjunct to the study of proxy data, researchers have employed an hierarchy of numerical models to study the LGM climate. The first approach employs atmosphere-only models with specified sea surface temperatures (SST). This was followed by atmosphere–mixed layer

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ocean models (e.g., Vettoretti et al. 2000). In the case of the atmosphere-only model, the surface air temperature (SAT) change is largely constrained by the specified LGM SSTs while the atmosphere–mixed layer models necessarily ignore changes in ocean circulation, heat transport, and their associated feedbacks. Coupled models of intermediate complexity, obtained by simplifying atmosphere and ocean representation, have been used to include some of these feedbacks (e.g., Ganopolski et al. 1998). However, the simplified representation of atmosphere–ocean feedbacks may or may not be appropriate for the large climate perturbation associated with the LGM. Three dimensional atmosphere–ocean coupled models represent our best effort to represent the physics of the climate system and should be the best tool with which to simulate the climate response to changing forcing and the climate of the LGM.

Coupled climate models are widely used to make projections of future warming in response to increasing greenhouse gas and other forcing (e.g., IPCC 2001 and references therein). There is a considerable range in projected warming due to differences in sensitivity between the various models. These future projections cannot yet be ‘verified’, and simulations of historical climate change provide some information on model fidelity. The instrumental record, which reliably spans only the past 150 years or so, is associated with rather modest changes in climate forcing and so is not a very powerful test of model sensitivity. Paleoclimate reconstructions offer an alternative testing strategy, one in which the climate forcing perturbation is large, but the evaluation is limited by uncertainties in the paleoclimate reconstructions. Nevertheless, climate model simulations of paleoclimate epochs can provide some measure of model sensitivity that is complimentary to the more recent historical comparisons.

There have been only a few attempts at LGM simulations with coupled models (i.e., models that include comprehensive three-dimensional representations of atmospheric and oceanic general circulation). Bush and Philander (1999) integrated the GFDL coupled model for 15 years after imposing LGM CO_2 and boundary

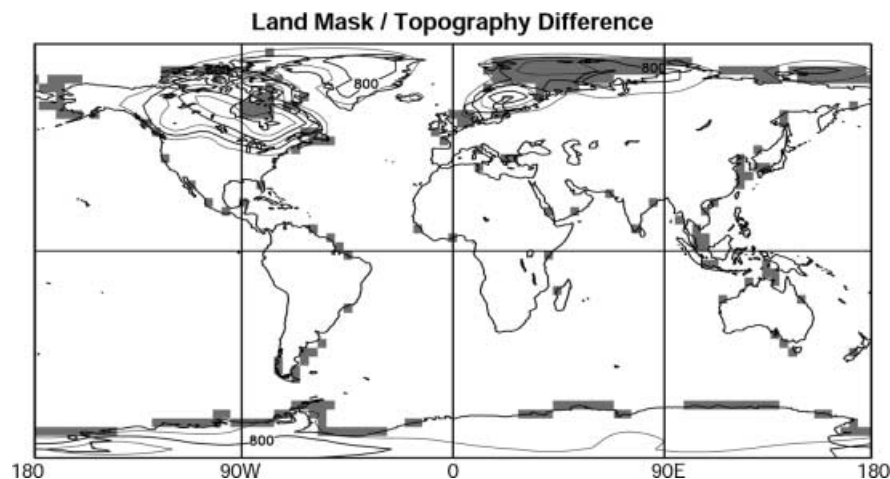
conditions. Their results, while clearly not representative of the equilibrium response, showed 4.3°C global mean cooling and $\sim 5\text{--}6^\circ\text{C}$ cooling over the tropical Pacific. Although not discussed, Bush and Philander (1999) obtained a slight warming in the Southern Ocean in their simulation. In a more recent LGM simulation using the Hadley Centre coupled model (HadCM3), after 770 years of integration Hewitt et al. (2001) found 3.8°C global mean cooling with about -1°C change over the tropical Pacific. They obtained anomalously warm temperatures over parts of the North Atlantic which they attribute to increased northward heat transport by a stronger North Atlantic overturning circulation. Finally, Kitoh et al. (2001) integrated the MRI coupled model under LGM conditions for 270 years. They obtained a global mean cooling of 3.9°C , with cooling over the tropics of 1.7°C , but patches of warming in the subtropical Pacific, and increased North Atlantic overturning. These LGM studies using coupled atmosphere–ocean models exhibit considerable variation in their results, and disagree in various respects with LGM reconstructions. For example, recent terrestrial and ocean geological proxy records such as snow line depression, tropical ice cores, and Barbados coral have shown that the tropical SAT and SST were more than 5°C lower than at present (Rind and Peteet 1985; Thompson et al. 1995; Guilderson et al. 1994). Moreover, the observed LGM North Atlantic overturning is inferred to have been substantially weaker and shallower than at present (Boyle and Keigwin 1987).

The objective of this study is to investigate the transient climate response to LGM conditions using the coupled atmosphere–ocean–sea ice model developed at Canadian Centre for Climate Modelling and Analysis (CCCma). In a companion paper, we analyze the near equilibrium response.

2 The coupled model

The second generation CCCma coupled climate model (CGCM2) is comprised of atmosphere, ocean, sea ice, and land surface

Fig. 1. Present-day and LGM land mask difference (*shading*) and ice-sheet topography change (*contours*)



components. It differs from an earlier version (CGCM1, see Flato et al. 2000) only in its representation of ocean mixing and sea-ice dynamics (Flato and Boer 2001). A detailed description of each component is given in other papers and so only a brief summary is provided here.

2.1 The atmosphere and land surface component

The atmospheric component of CGCM2 is the second generation atmospheric GCM described by McFarlane et al. (1992). This model employs a triangular spectral truncation with 32 longitudinal wave numbers (T32), corresponding to a surface grid spacing of approximately 3.75° in latitude and longitude. It has 10 vertical levels with a hybrid vertical coordinate.

Specific humidity is the prognostic moisture variable in this model and an interactive cloud scheme is used. The fractional cloud

cover is evaluated from the prognostic moisture variable through a relative humidity threshold, which is prescribed as a function of atmospheric pressure from 85% to about 90%. Whenever the local relative humidity is supersaturated, excess water vapour condenses and precipitates. Moist convective processes are parameterized by a convective adjustment scheme. Ozone is specified from observations. The effects of clouds, water vapor, carbon dioxide, oxygen, and ozone are included in the calculation of solar and terrestrial radiation following Fouquart and Bonnel (1980) and Morcrette (1984). In order to properly represent the diurnal cycle, the radiative processes are evaluated several times per day. The solar constant is specified to be 1370 W m^{-2} . Orographic gravity wave drag is according to McFarlane (1987).

The force-restore method is used to calculate the surface energy balance. Soil moisture is represented in terms of a simple bucket model, but with regionally varying field capacity depending on vegetation and soil characteristics. The surface albedo is specified depending on the surface type. Over the land it varies with the 24 specified vegetation categories of the Wilson and Henderson-Sellers (1985) dataset. Where the land surface is covered with snow, the surface albedo varies from 0.55 to 0.9 depending on a prognostic snow age factor and snow masking depth. Over the open ocean the albedo is specified as a function of latitude from 0.06 in the tropics and 0.17 in the mid- and high-latitudes, over sea ice it is parameterized as a function of surface temperature. Where there is a partial snow or sea-ice cover, the albedo is represented as a linear combination of the relevant surface albedos.

Freshwater flux into the ocean from the land is included. The scheme involves instantaneous transfer of runoff from land to the ocean using specified river drainage basins which have been modified to reflect the altered land distribution and ice sheet topography of the LGM.

2.2 The oceanic and sea-ice component

The oceanic component of the CGCM2 is the GFDL MOM version 1.1 (Pacanowski et al. 1993). The horizontal grid spacing is 1.875° in latitude and longitude, such that four grid cells corresponds to one atmospheric grid cell. The model has 29 vertical layers, the finest resolution being in the upper four layers whose thickness is 50 m.

The ocean model includes realistic bottom topography from Gates and Nelson (1975) with slight modifications in the Greenland–Iceland–Norwegian seas to better reproduce overflow. Exchanges across the Canadian Arctic Archipelago and the strait of Gibraltar are represented by horizontal mixing. Flow through Bering Strait is resolved.

Ocean mixing is represented by vertical and isopycnal diffusion, along with the eddy-stirring parameterization of Gent and McWilliams (1990) (GM). Isopycnal diffusivity is $1000 \text{ m}^2 \text{ s}^{-1}$ with a background horizontal diffusivity of $100 \text{ m}^2 \text{ s}^{-1}$. Vertical diffusivity is $3 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$. When the density stratification is unstable, convective adjustment is performed by increasing the vertical diffusivity following Bryan and Lewis (1979).

CGCM2 includes a dynamic-thermodynamic sea-ice model. Sea-ice dynamics, not included in the earlier version (CGCM1), employ the cavitating fluid rheology of Flato and Hibler (1992). Heat transfer through the ice is represented by a slightly modified version of the 0-layer scheme of Semtner (1976).

2.3 Spin-up and coupling

The atmospheric model, coupled to a slab mixed-layer ocean component (McFarlane et al. 1992) is integrated for 30 years to obtain estimates of heat, freshwater and momentum fluxes. The ocean model is then run for approximately 4000 years using these fluxes plus a weak restoring to climatological SST and sea surface salinity (SSS). The atmosphere and ocean components are then coupled and integrated for an additional 50 years with restoring terms included. Monthly heat and freshwater flux adjustment fields were then computed by averaging these restoring terms over the

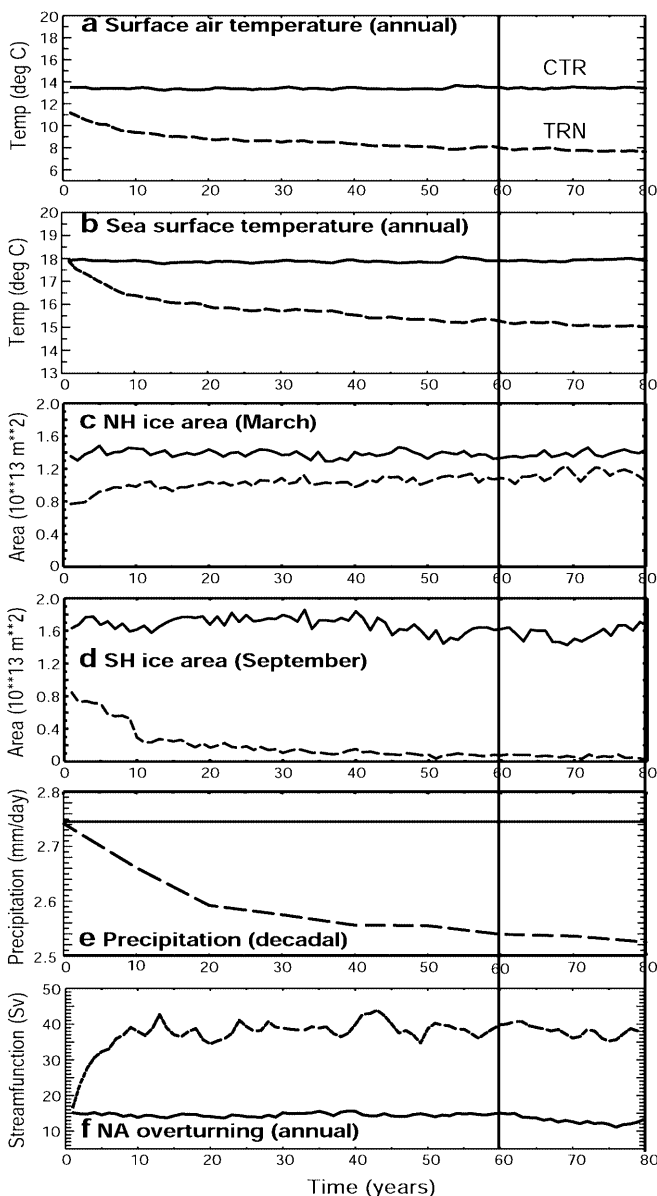


Fig. 2a–f. Time series of annual-mean, **a** surface air temperature, **b** sea surface temperature, **c** March NH ice area, **d** September SH ice area, **e** decadal-mean precipitation, **f** annual-mean North Atlantic overturning stream function

last 30 years of this integration. These flux adjustment fields were used in all subsequent model simulations. Although the use of flux adjustment is not ideal, it produces a control climate which is in reasonable accord with observations (Flato et al. 2000; Lambert and Boer 2001).

The atmosphere and ocean component interacts once per day exchanging heat, fresh water, and momentum. The coupling procedure of this model is basically the same as that of CGCM1 (Flato et al. 2000) except that the ice dynamics component is included and runs with a 1 day time step.

3 Experiments

Two experiments are analyzed in this study. The control experiment, hereafter referred to as CTR, has a specified CO₂ concentration of 330 ppm, and a contemporary land mask and topography. The second experiment involves conditions representative of the LGM but, because we analyze only the initial transient evolution toward an LGM climate, we refer to this experiment as TRN. The imposed LGM conditions are as follows:

1. The land mask is modified to account for the lower sea level (approximately 120 m) following Fairbanks (1989) as depicted by shading in Fig. 1. This expands the continental margin and causes closure of the Bering Strait.
2. LGM ice sheet topography is imposed following the reconstruction of Peltier (1994) (Fig. 1). Because of the smoothness of the imposed ice sheet, the gravity wave drag no longer operates over the region.
3. The CO₂ concentration of the atmosphere is reduced to 235 ppm from the 330 ppm which is used in our control integration. The difference of CO₂ concentration between the two experiments gives approximately the same CO₂ forcing change

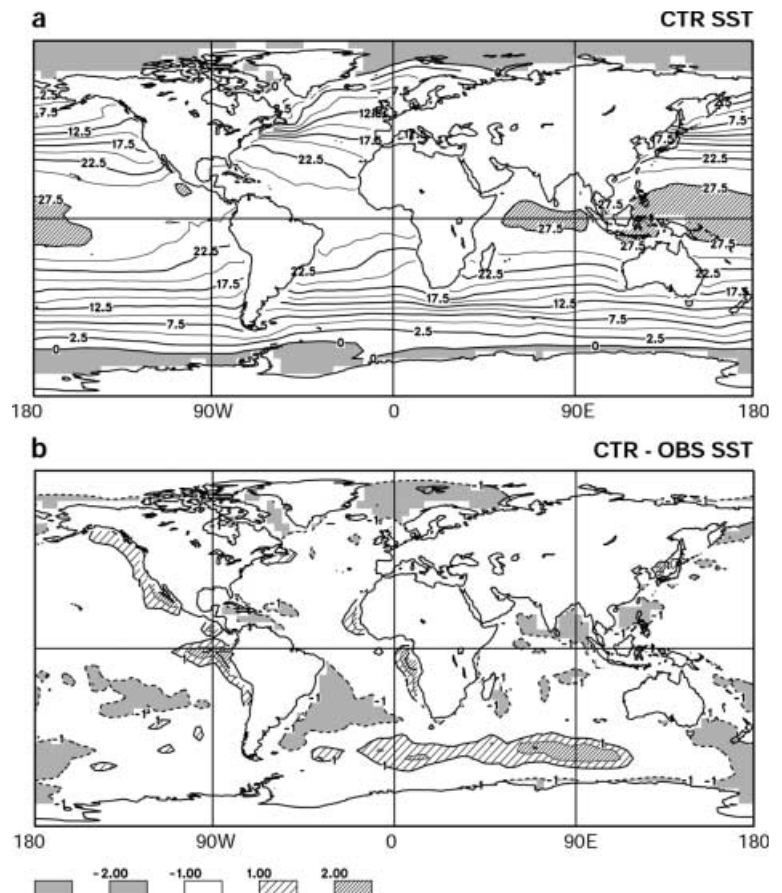
as that due to the difference between the observed LGM (approximately 200 ppm from Barnola et al. 1987) and the pre-industrial (280 ppm) concentrations.

4. Orbital parameters are unchanged since, during the LGM (~21 kaBP) the periods of obliquity (41 ka) and precession (23 and 19 ka) are almost the same as at present and the eccentricity (with period 100 ka) is also basically unchanged. Several model results have shown that the radiative forcing change induced by LGM orbital parameters is indeed negligible (e.g., Hewitt and Mitchell 1997; Broccoli 2000).
5. Vegetation and soil types are unchanged except over glaciated surfaces. Land points arising due to sea-level reduction are assigned a medium soil type and albedo specification.

4 Simulated climate

The time evolution of global mean surface quantities in the CTR (solid) and TRN (dashed) simulations is shown in Fig. 2. The CTR simulation represents an equilibrium climate state as indicated by the lack of trends in temperatures. LGM conditions are imposed on this climate equilibrium and the modelled climate adjusts strongly to this change of forcing during the first decade and then evolves more slowly towards the LGM equilibrium. The modification of LGM-based land mask (Fig. 1) is responsible for the initial mismatch of global-mean values between CTR and TRN. For example, in both hemispheres the TRN sea ice area is initially reduced by about 50% where large areas of the continental shelves in the Arctic and the Antarctic become land.

Fig. 3. **a** Horizontal distribution of annual-mean sea surface temperature simulated in CTR. **b** Difference between the simulated sea surface temperature and observed (Levitus and Boyer 1994). Units °C



After 50 years of simulation, most surface quantities have attained a “quasi-equilibrium” and the system then evolves more slowly toward an LGM equilibrium. (This slow adjustment continues for at least another 1000 years). We concentrate here on results over the first 80 years following the imposition of the LGM conditions focusing on the mechanisms of climate response to the LGM change in forcing. Near equilibrium results and comparisons to proxy data are provided in a companion paper. We provide also an evaluation of the CGCM2 control run so as to set the stage for the analysis of the evolution toward LGM conditions in the TRN experiment. Because the atmospheric results are similar to those presented earlier by McFarlane et al. (1992) and Flato et al. (2000) and because the long time scale climate is controlled by the ocean, we focus primarily on oceanic thermohaline properties, circulation, and sea-ice distribution. Except where noted, we display annual mean results averaged over a 20 year segment (60–80 years) as depicted in Fig. 2.

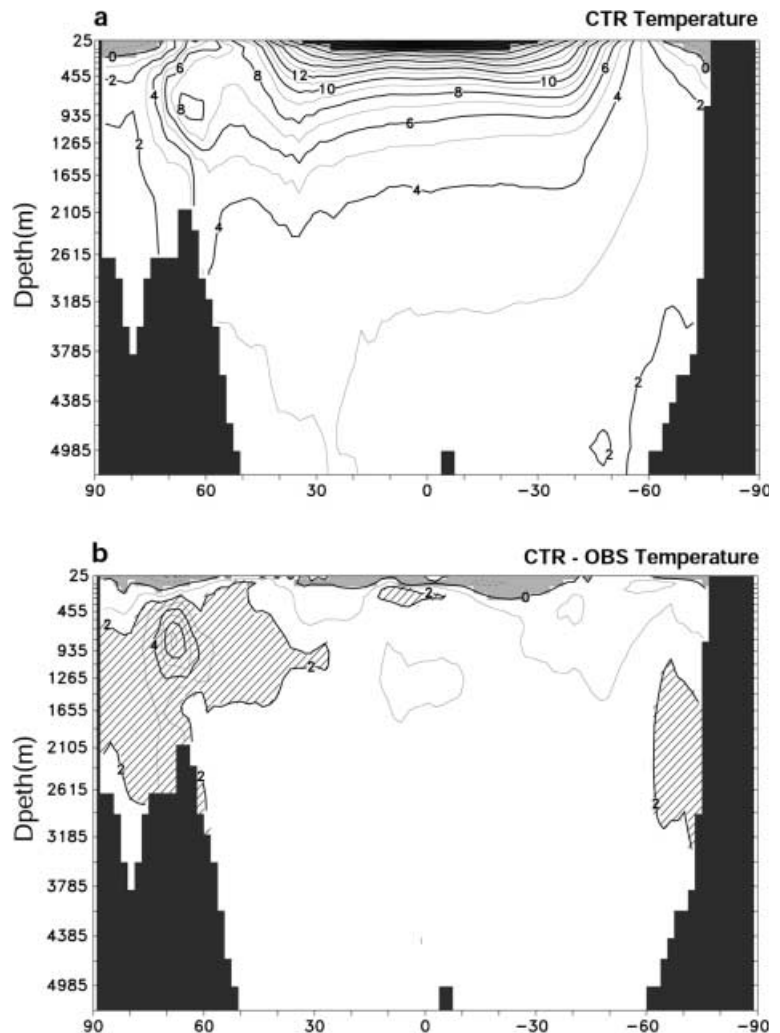
4.1 Temperature

4.1.1 Control temperature

The geographic distribution of simulated SST and its difference from the observational estimate of Levitus and Boyer (1994) are shown in Fig. 3. Not surprisingly, since the model is flux corrected, CGCM2 simulates the horizontal distribution of SST reasonably well, with difference of less than 1 °C except in a few small patches. This represents some improvement over the control simulation of CGCM1 (Flato et al. 2000) particularly in the northern North Atlantic.

Figure 4 presents the zonally averaged global ocean temperature simulated by CGCM2 and its difference from observations (Levitus and Boyer 1994). The intermediate temperature is better simulated than was the case with CGCM1, and this is attributed to the effects of the isopycnal and GM mixing scheme. Intermediate and deep water at high-latitudes, though improved over CGCM1, remain warmer than observed.

Fig. 4. **a** Zonally averaged annual-mean ocean temperature simulated in CTR. **b** Difference between the simulated temperature and observed (Levitus and Boyer 1994)



4.1.2 Transient temperature response

In response to the imposition of the LGM conditions described in Sect. 4, the global SAT and SST decrease with time and after 80 years SAT and SST decrease by 5 °C and 3 °C, respectively (Fig. 2). The evolution of the zonally averaged decadal-mean surface SAT and SST is shown in Fig. 5. SAT decreases rapidly in high northern latitudes, less rapidly in the tropical regions, and there is an unexpected warming between 50° and 70°S. The strong inter-hemispheric asymmetry in the temperature response is a marked feature of the initial adjustment to LGM conditions. The strong cooling in the northern high latitudes is largely due to the change in forcing associated with the high-elevation glaciated surface over northern North America and northern Europe (see Fig. 1). The southern hemisphere (SH) low latitude cooling is slower, and the high latitude warming is as-

sociated with the ocean as seen in Fig. 5b. In comparison to the SAT change, the SST change is rather symmetric latitudinally and a similar although more modest SST increase is seen around 70°N. The mechanism responsible for the transient localized warming response to LGM boundary conditions is discussed later.

The latitude-height temperature change for the atmosphere and ocean is plotted in Fig. 6. In the atmosphere, the largest cooling is found in the northern polar region, and in the tropical upper troposphere. Less cooling is found at southern high latitudes in the atmosphere and warming is seen near the surface. The pattern of temperature change in the atmosphere is similar in pattern but opposite in sign to that associated with greenhouse gas warming in the model (Boer et al. 2000b). In the ocean, cooling is limited to the upper few hundred meters at low and mid latitudes, but extends to

Fig. 5a,b. Temporal evolution of the difference between CTR and TRN in zonally averaged decadal-mean, **a** surface air temperature, **b** sea surface temperature

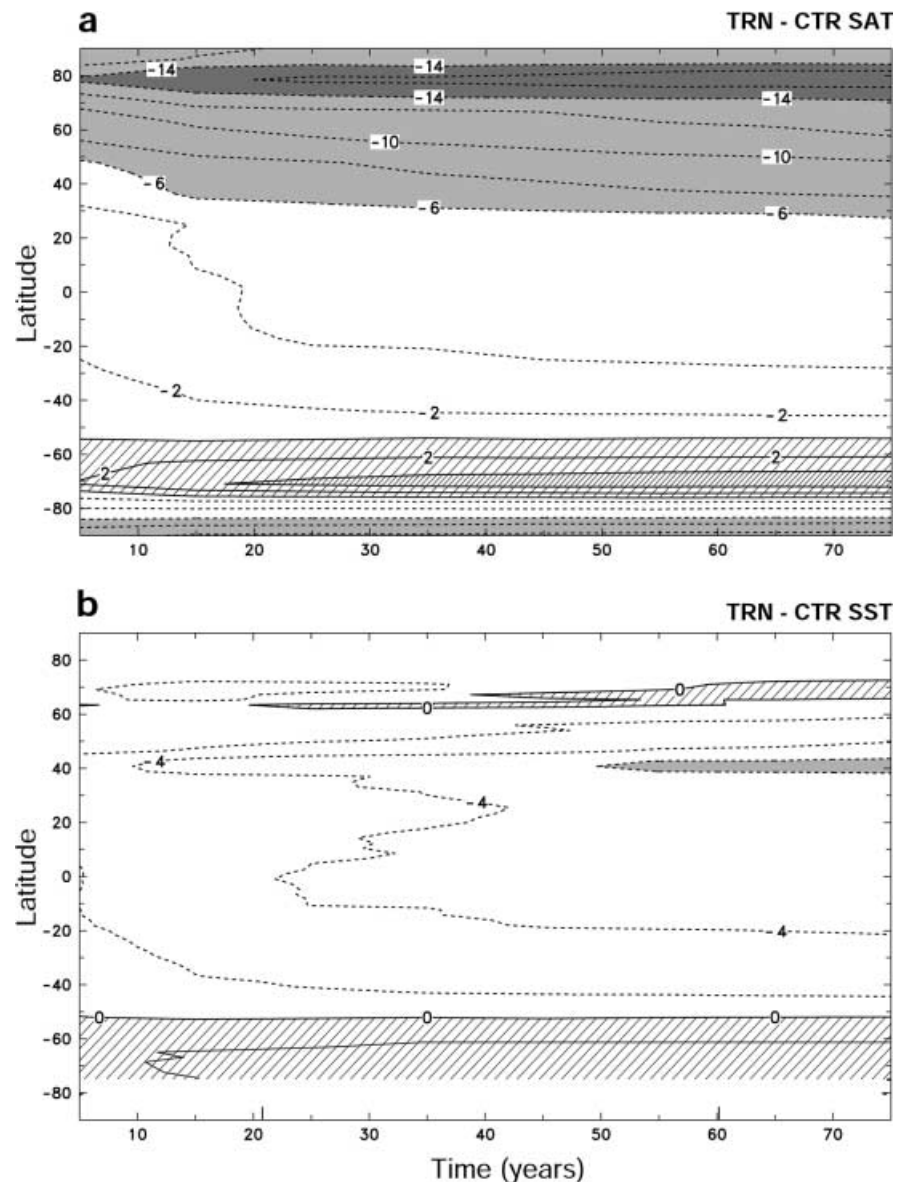
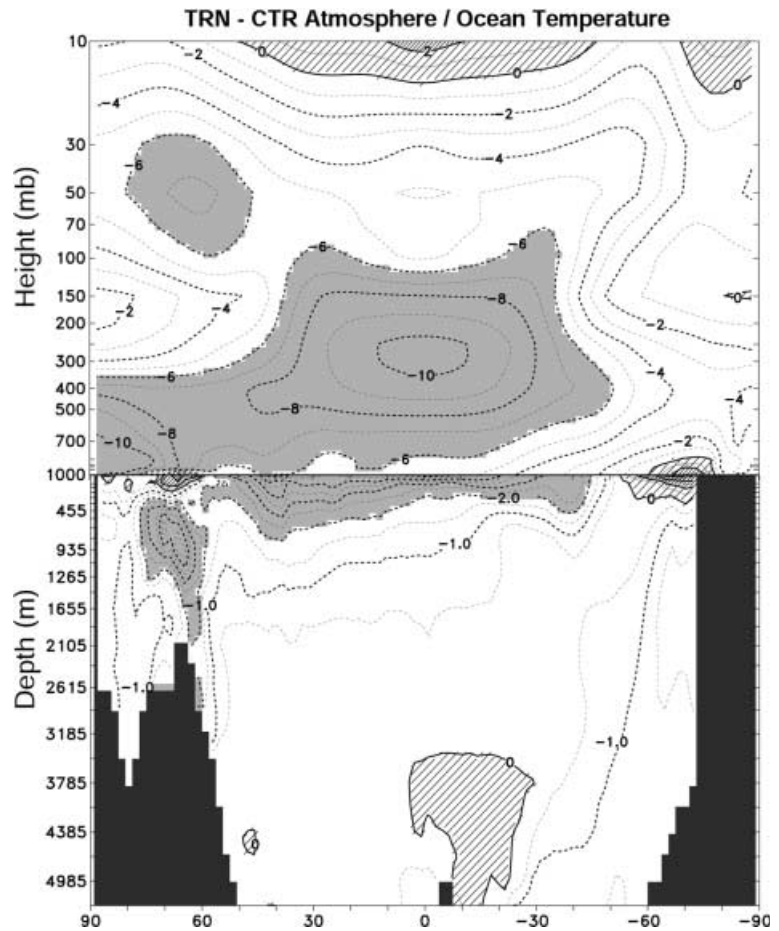


Fig. 6. Zonally averaged annual-mean atmosphere and ocean temperature difference between CTR and TRN



the ocean bottom at high latitudes where it is overlain by waters which have warmed, the signature of enhanced deep convection. This LGM cooling case contrasts with the greenhouse gas warming case (Flato and Boer 2001) in which the warm anomaly penetrates to much shallower depths, less than 2000 m in both hemispheres. A similar result has been obtained by gradually increasing and decreasing CO_2 (Manabe et al. 1991). The stronger (weaker) convection in the cooling (warming) case leads to the deeper (shallower) penetration of the cold (warm) anomaly.

Figure 7a gives the geographical distribution of the SAT difference between TRN and CTR. A cooling of more than 20°C is apparent over the glaciated areas of North America and northern Europe due to the increase in elevation and albedo accompanying the introduction of the ice-sheet boundary conditions. In addition to the inter-hemispheric asymmetry seen in Fig. 5a, there is a very strong land-sea contrast with stronger cooling over land than ocean. The strong warming of more than 10°C around Antarctica is a striking feature. This Southern Ocean temperature response is not a feature of LGM simulations using atmosphere-mixed layer ocean models, including the CCCma model (Joussaume and Taylor 2000).

Figure 7b displays the SST difference between TRN and CTR. Local ocean temperature minima (6°C cooling

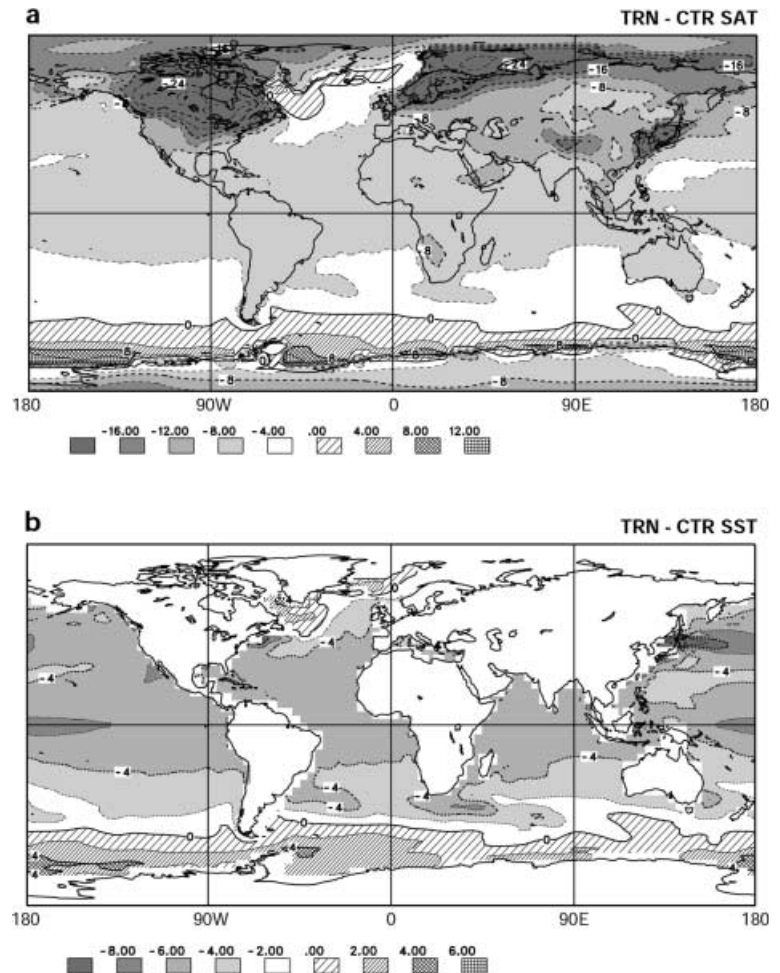
or more) appear in the north-western Pacific and Atlantic at about 40°N and in the western equatorial Pacific. The locations of extratropical cooling coincide with locations of enhanced offshore winds, whereas the equatorial cooling is connected with intensified trade winds and enhanced upwelling (see Sect. 4.3). The transient warming in Fig. 5b is seen to occur around Antarctica and in the Labrador and Greenland seas in Fig. 7b.

4.1.3 Anomalous warming

We investigate the evolution of the localized warming in more detail focusing on the Weddell Sea ($20\text{--}50^\circ\text{W}$ and $65\text{--}75^\circ\text{S}$) which is a typical region of strong transient warming (see Fig. 7). Figure 8a illustrates the time evolution of annual-mean differences (TRN minus CTR) of SAT, sea-ice thickness, SSS, and surface potential density (σ), all averaged over the Weddell Sea. Figure 8b shows a time-depth plot of the change in ocean temperature while Fig. 8c displays the annual-mean frequency of oceanic convection (total number of convection events/total number of time steps) for the TRN run.

For the first few years after imposing glacial boundary conditions, SAT begins to cool by about 1.5°C and in turn ocean surface temperatures decrease by about 0.5°C . In conjunction with this cooling, SSS increases

Fig. 7a,b. Horizontal distribution of annual-mean difference between CTR and TRN in **a** surface air temperature, **b** sea surface temperature



largely due to increased sea-ice formation and associated brine release. This cooling and salinification increases surface potential density and eventually (by year 4) leads to an abrupt increase in the frequency of deep convection which exchanges heat between the cool ocean surface layer and the warmer deep water. The increased convective activity warms the upper 1000 m while cooling the ocean below this depth. Sea-ice thickness and coverage decrease and this is reinforced via the positive ice-albedo feedback. Thus the initial cooling and growth of sea ice stimulates its self destruction in a manner similar to that described by Martinson (1990) using an analytical model and by Gordon and Huber (1990) through observation in the north-eastern Weddell Sea. Although initially associated with changes in ice thickness, the continuing increase in SSS relies on vertical mixing of saline deep water ($S > 34.7$ psu) and on increased evaporation due to anomalous warming (see following subsection).

4.1.4 Sea ice and ocean convection

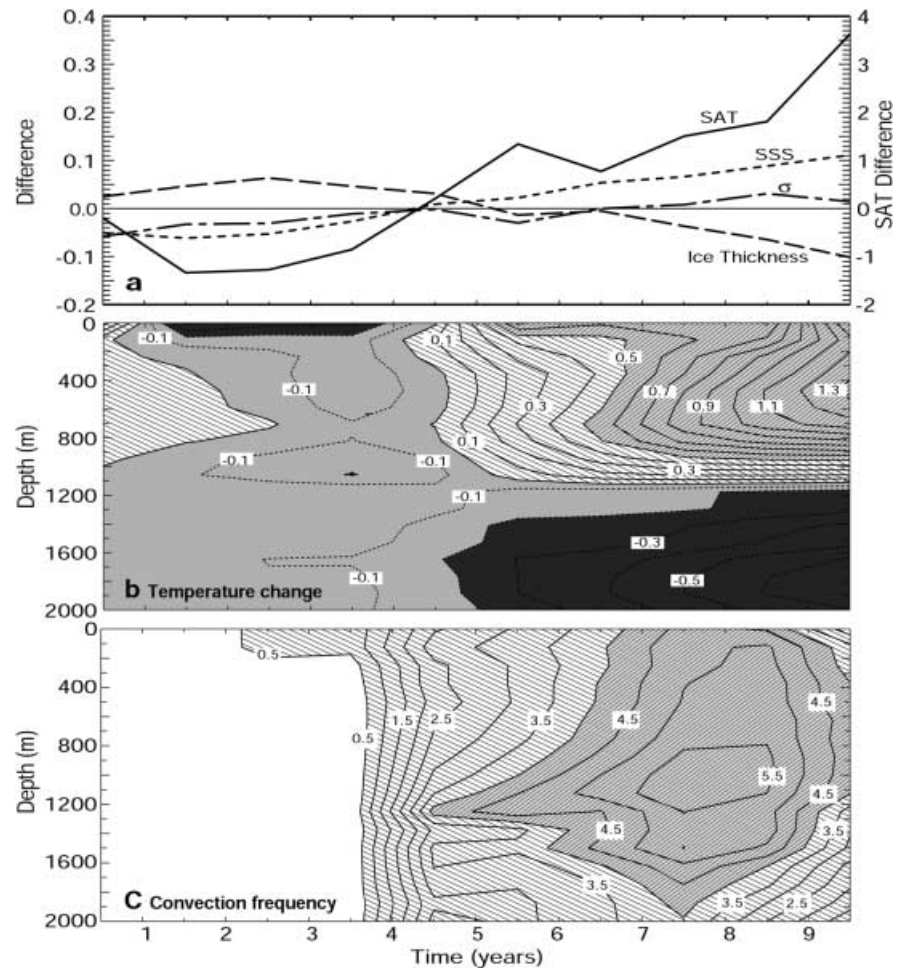
The sea-ice distribution is an important climatic determinant through the albedo feedback mechanism and

through ice effects on atmosphere–ocean heat flux. Sea-ice melting and freezing also influences the rate of local oceanic convection and subsequently the rate of deep and bottom water formation. In the northern hemisphere (NH), sea ice covers the entire Arctic Ocean and its boundary in the Greenland Sea is largely controlled by advection of warm water from the North Atlantic. In the SH, heat supplied by vertical mixing plays a critical role in determining sea ice extent. The observed winter maximum sea ice extent is estimated to be about 15.7×10^6 km² in the NH (and occurs in late March) whereas the SH maximum (in September) is 19×10^6 km² (Gloerson et al. 1992).

The mean March and September ice thickness fields produced in CTR and TRN are shown in Fig. 9 for the NH and SH. The annual-mean frequency of convection at 500 m is also shown by the contour lines in Fig. 9. Although the NH ice cover is somewhat thin in CTR, its extent (14×10^6 km²) is close to that observed. Ice thickness in the SH is even more poorly known than in the NH, but the thickness produced by CGCM2 is certainly plausible although the September ice extent (16×10^6 km²) is somewhat less than observed.

With the imposition of LGM conditions, the NH winter sea-ice area is initially reduced due to the change

Fig. 8. **a** Temporal evolution of annual-mean difference between CTR and TRN averaged in the Weddell Sea (20–50°W and 65–75°S) for surface air temperature (SAT) in °C, sea surface salinity (SSS) in psu, sea-ice thickness in meters, and surface potential density (σ) in kg m^{-3} . SAT changes refer to the *right-hand scale*. **b, c** Time-depth evolution of annual-mean, **b** difference between CTR and TRN in ocean temperature, **c** convection frequency simulated in TRN. Units °C and percent



in the land area as depicted by hatching in Fig. 9b and 9d. Ice cover thereafter slowly increases with time although its area remains smaller than CTR after 80 years (Figs. 2c and 9b). In the central Arctic, the TRN winter sea ice is much thicker than that of CTR and in the North Pacific it extends farther south. In contrast to the NH, the SH winter sea ice decreases rapidly and almost disappears in 80 years (Figs. 2d and 9d). The virtual disappearance of sea ice around Antarctica is due primarily to the increased upward oceanic heat flux as a result of the intensification of convection and secondarily to the increased southward ocean heat transport described later.

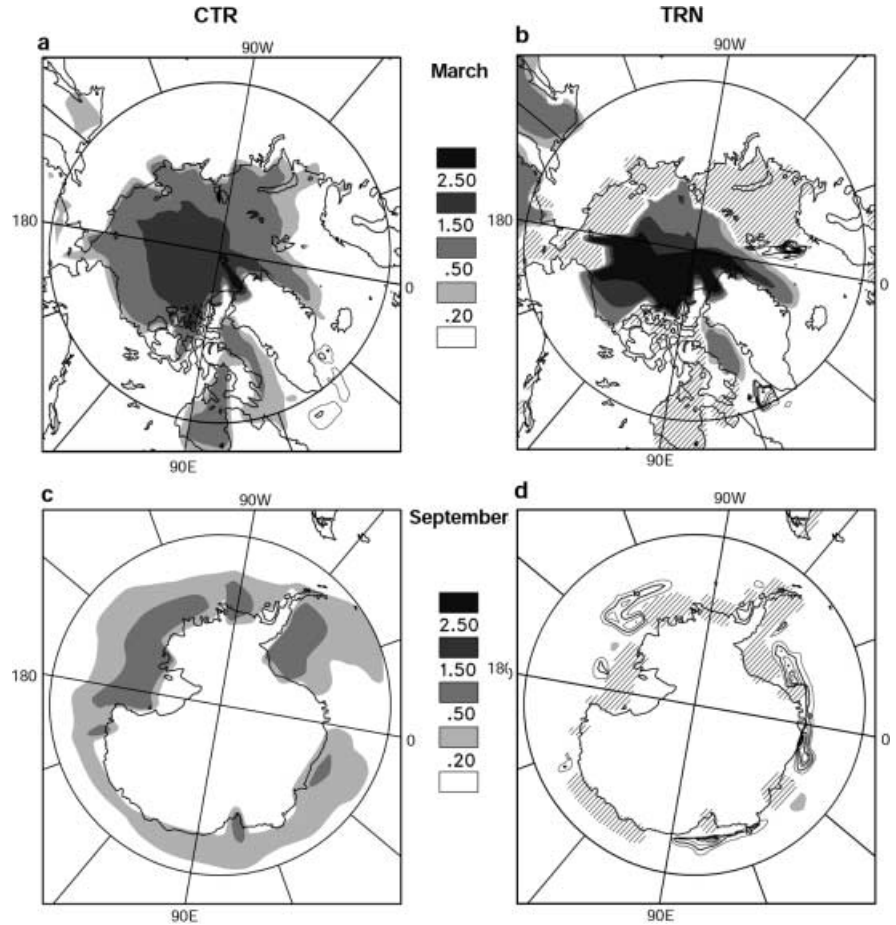
The convective activity in the control run which uses the isopycnal/GM mixing scheme is thought to be more realistic than that with the previous horizontal mixing formulation, especially in the Southern Ocean (Fig. 9a, c) (see also Danabasoglu and McWilliams 1995). With glacial conditions, convection becomes more vigorous over much of the Southern Ocean and in the Greenland and Labrador Seas. These areas are broadly coincident with the locations of transient warming shown in Fig. 7 and exhibit behavior similar to that seen in the Weddell Sea (Fig. 8).

4.1.5 Surface heat balance

Figure 10 shows the geographic distribution of differences in surface radiative, turbulent (latent plus sensible), and net heat fluxes. Positive (negative) values indicate that the change is such as to warm (cool) the surface. Over North America and northern Europe, the change in radiative flux acts to cool the surface primarily due to the reduction of shortwave radiation input caused by the increased surface albedo over the ice sheet, whereas turbulent heat fluxes act to warm the surface due to reduced evaporation and sensible heat exchange. The reduction in shortwave heating over land is offset by the reduction in cooling by the turbulent heat fluxes, reflecting the new energy balance at a colder temperature.

The situation is very different in the ocean, especially where the anomalous warming occurs. In the Southern Ocean, the radiative change acts to warm the surface slightly due to increased shortwave absorption associated with the reduction of sea ice, whereas the turbulent heat flux changes act to cool ocean due to increased evaporation and sensible heat flux associated with the surface warming. In the Greenland and Labrador seas, the

Fig. 9a–d. Horizontal distribution of sea ice thickness in meters (*shading*), Northern Hemisphere March for, **a** CTR, **b** TRN; Southern Hemisphere September for, **c** CTR, **d** TRN. *Contour lines* represent annual-mean oceanic convection frequency at 500 m simulated in CTR and TRN. The areas that became land in TRN are represented as *hatching*. Units meters and percent



turbulent heat flux change acts to strongly cool the surface (by more than 100 W m^{-2}) due to a combination of warmer surface waters, colder air advected off of the adjacent (ice-covered) land, and stronger winds. The slight negative radiative heat flux change over the northern high-latitude oceans is due to increased surface albedo associated with slightly increased Arctic sea ice. At lower latitudes, the biggest change is in the latent heat flux acting to warm the surface and represents the decrease in evaporation associated with a decrease in SST.

Though a very strong local net heat flux change is evident in the NH convection regions as seen in Fig. 10c, the area involved is relatively small and its contribution to the total heat flux is smaller than that in the Southern Ocean. This is illustrated in the zonally integrated differences in surface heat flux components evaluated over latitude bands as

$$\Delta \hat{F} = a^2 \cos \varphi \Delta \varphi \int_0^{2\pi} \Delta F d\lambda.$$

The area under the curves in Fig. 11 represents the actual heat flux in that latitude band. At low latitudes over land the net heat flux is close to zero representing the rapid adjustment of the heat balance over land; the non-zero values in mid to high latitudes are due to the energy used to melt or freeze the snow and ice. In the

low latitude oceans, the large changes in latent heat flux (less obvious in the previous figure) acts to warm the surface, although it is roughly offset by both radiative and sensible heat cooling.

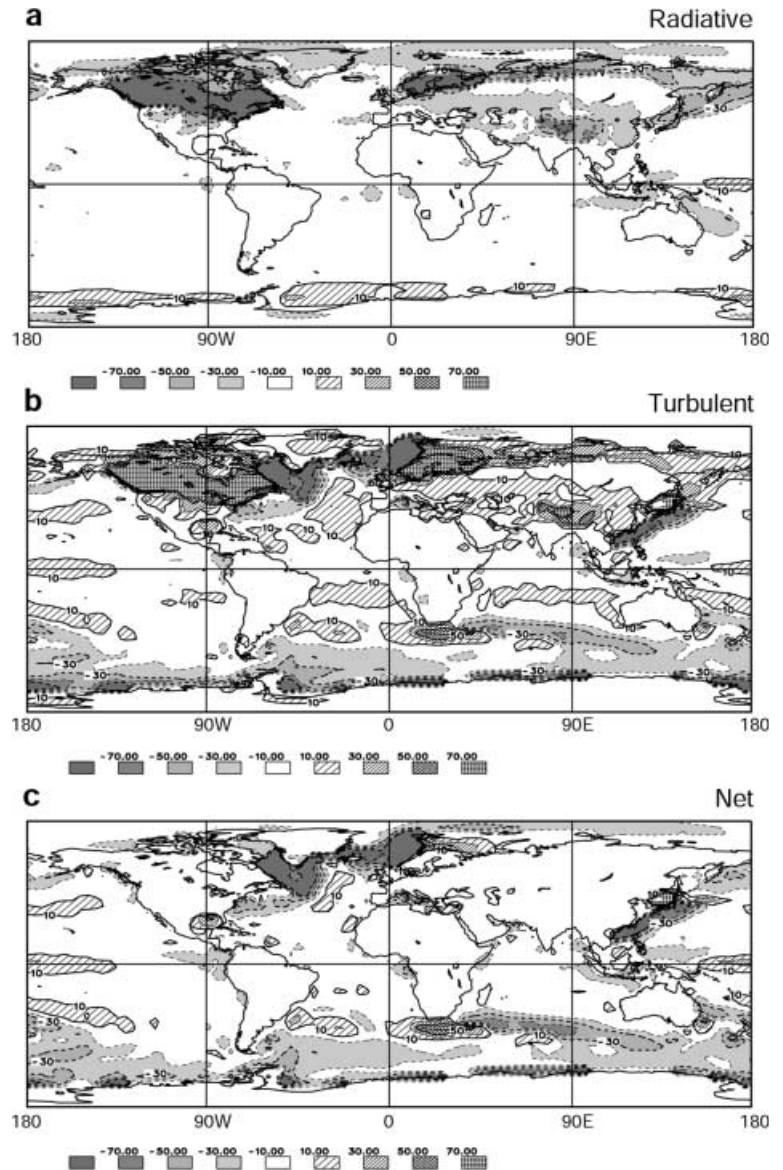
Overall, the surface heat is in approximate balance over the low-latitude land and oceans in the transient phase, whereas in the high-latitude oceanic convection regions the large amount of surface heat, which is transferred from the deep ocean via vigorous vertical mixing and also by the stronger poleward heat transport associated with the intensified overturning circulation (see Sect. 4.3), is lost to the atmosphere by very active turbulent heat fluxes. By this process, the ocean, including the deep ocean continuously loses heat and cools with time.

4.2 Salinity

4.2.1 Control salinity

The spatial distribution of SSS, shown in Fig. 12, agrees well with the observational estimate of Levitus et al. (1994) as expected since the model is flux corrected. The new mixing scheme in CGCM2 increases the zonally averaged salinity considerably compared to CGCM1

Fig. 10a–c. Horizontal distribution of the change in annual-mean, **a** radiative, **b** turbulent, **c** net heat fluxes. Units W m^{-2}



(Fig. 13). The low salinity Antarctic Intermediate Water (AAIW) tongue is much more pronounced in CGCM2 than was the case in CGCM1. Second, in CGCM2 the deep ocean salinity is much closer to observations (it was generally too fresh in CGCM1). Moreover, the saline tongue in the Southern Ocean associated with the North Atlantic Deep Water (NADW) intrusion is better represented (Fig. 13a). The weaker Southern Ocean convection associated with the isopycnal/GM mixing parameterization used in CGCM2 is responsible for the improvement of the low salinity Antarctic water tongue and allows warm and saline NADW to extend farther south, leading to increased global deep ocean salinity.

4.2.2 Hydrological cycle and salinity response

Imposition of LGM conditions substantially modifies the global hydrological cycle. Associated with the

decreased surface temperature, the global hydrological cycle weakens, and global mean precipitation and evaporation decrease rapidly during the first 20 years and by 80 years have decreased by 8% (Fig. 2). Figure 14 shows time-dependent zonal-mean precipitation (P), evaporation (E), E minus P , and SSS change. The imposed glacial conditions lead to a general decrease in precipitation, especially over the latitude bands of the NH ice-covered region, the tropics, and the southern high latitudes. Evaporation decreases as well except for the southern high latitudes where it increases substantially due to the anomalous surface ocean warming and the retreat of the ice cover. The change in E minus P in turn affects the SSS change. In the southern high latitudes, evaporation largely exceeds precipitation, contributing to the increase of SSS. A similar pattern is seen in the northern mid-latitudes around 40°N , but with opposite sign. A rapid and large increase in SSS with time is seen in the northern high

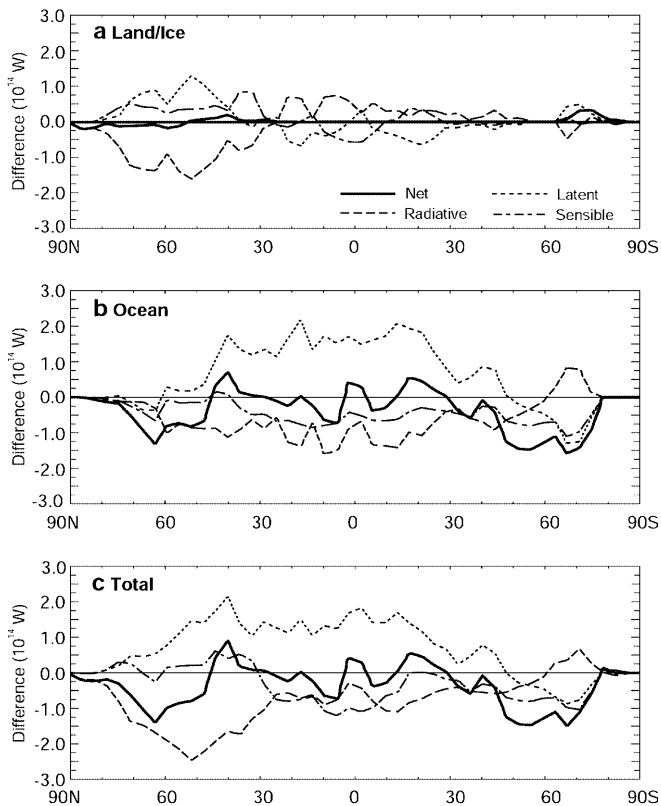


Fig. 11a–c. Latitudinal profile of zonally integrated heat component difference between CTR and TRN over, **a** land and ice, **b** ocean, and **c** entire globe. Units 10^{14} W

latitudes in spite of the slight excess of evaporation over precipitation. This is largely due to the change in the amount of river runoff as explained below.

Figure 15 shows zonally averaged salinity difference between TRN and CTR. In the Arctic Ocean, the salinity increase is not only limited to the surface layer but extends to the bottom as a result of enhanced vertical mixing. In the Southern Ocean, the increased surface salinity discussed earlier extends to a depth of about 1000 m, eroding the upper ocean stratification apparent in Fig. 13a. This enhanced salinity signal is carried northward along the AAIW core via isopycnal mixing.

Figure 16a displays the spatial pattern of E minus P change between CTR and TRN. The pattern is broadly similar to that obtained under global warming (Boer et al. 2000b), but with opposite sign. Of particular note is the westward shift of precipitation in the tropical western Pacific. This together with the westward extension of the cold tongue (see Fig. 7b) and intensified trade winds (see next section) in the tropical Pacific indicates a tropical

Fig. 12. **a** Horizontal distribution of annual-mean sea surface salinity simulated in CTR. **b** Difference between the simulated sea surface salinity and observed (Levitus et al. 1994). Units psu

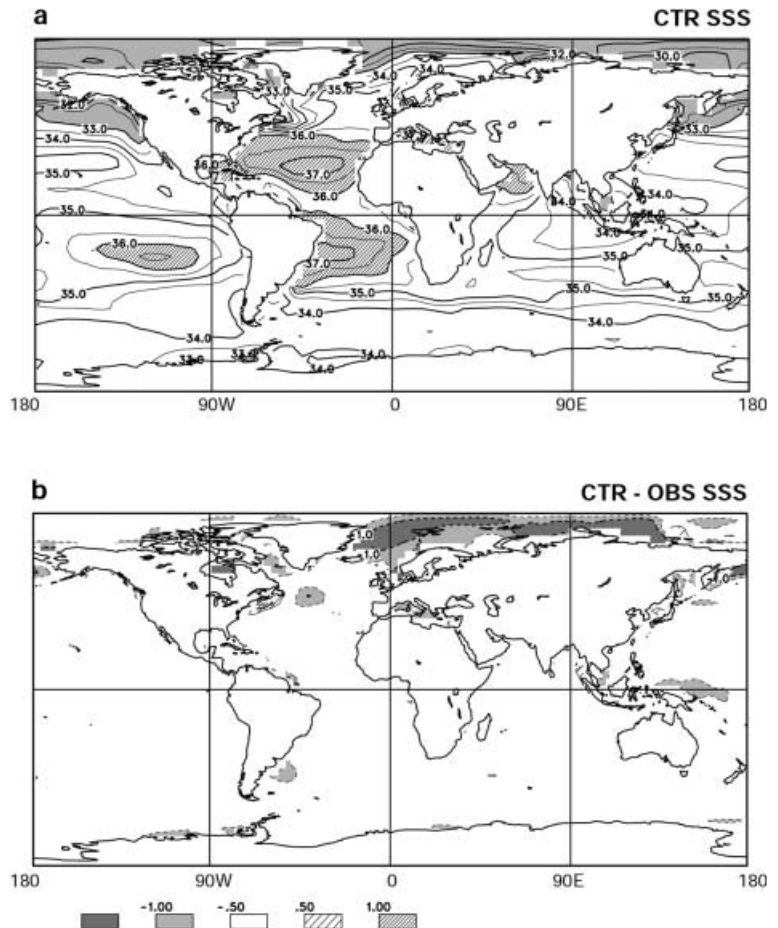
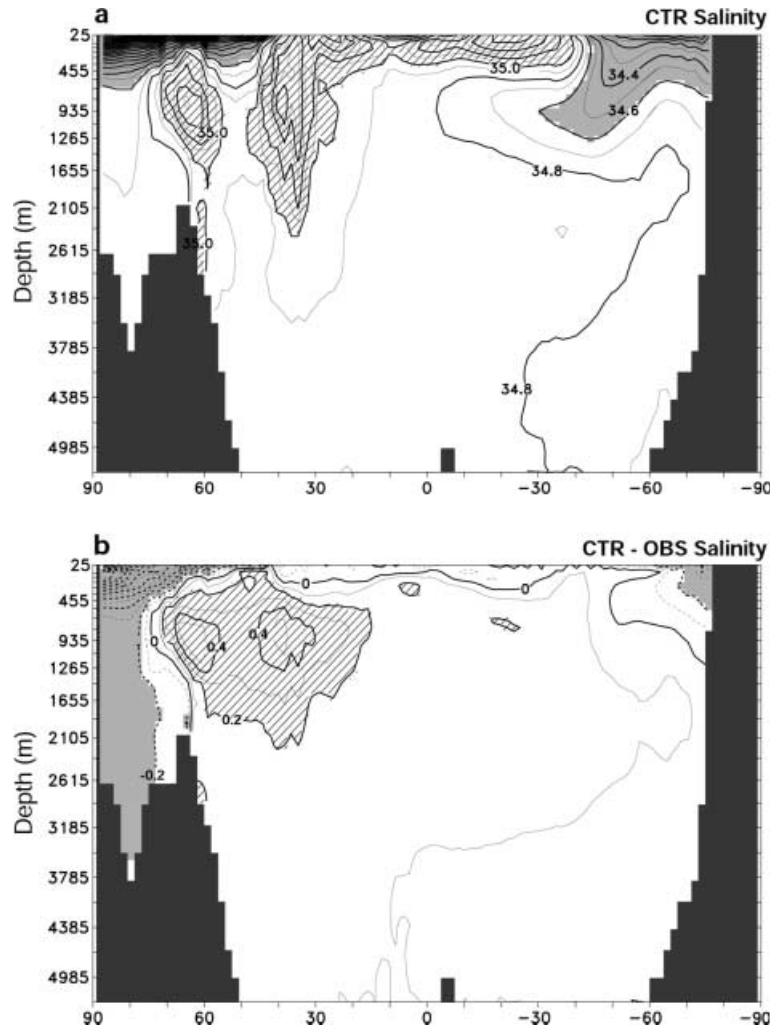


Fig. 13. **a** Zonally averaged annual-mean salinity simulated in CTR. **b** Difference between the simulated salinity and observed (Levitus et al. 1994)



La Niña-like mean response under glacial conditions. The same mechanisms invoked to explain the El Niño-like response under warming conditions (Yu and Boer 2001), operate in the opposite phase to produce a La Niña in a colder climate in the coupled model.

Figure 16b shows the corresponding changes in SSS between the two experiments. The change in the local evaporation minus precipitation broadly contributes to the SSS change as shown in Fig. 16. Additionally, the change in the amount of river discharge, the convective mixing, and the horizontal advection affects on the regional SSS change. Major regional features are explained as follows:

1. Arctic Ocean: the decreased river runoff to the Arctic (by about half of that of the CTR, Table 1), due to colder and hence drier climate and the accumulation of snow over the ice sheets, is mainly responsible for the SSS increase. Additionally, the increased convective mixing contributes to the SSS change.
2. North Atlantic: increased river runoff from the Mississippi and the Amazon rivers (Table 1) and increased horizontal advection of fresh water by the Gulf Stream freshens the North Atlantic. The

increase in Mississippi runoff is due to the increase of precipitation over evaporation south of the Laurentide ice sheet. Increased Amazon runoff is due to the overall wetter climate in the northern South America (Fig. 16a)

3. Southern Ocean: the convective mixing of relatively fresh surface water (34.0 psu) with the more saline deep water (34.8 psu) (see Figs. 12, 13) together with the increase in freshwater flux ($E - P$) is mainly responsible for the increase in SSS.

4.3 Ocean circulation

4.3.1 Zonal-mean sea-level pressure, surface wind, and wind stress

Figure 17 shows the change in zonally averaged mean sea level pressure, surface wind, and zonal surface wind stress. Associated with the change in the vertical distribution of temperature in the atmosphere under glacial conditions (Fig. 6), the mean sea level pressure (MSLP) generally increases at high latitudes. In the SH, the polar increase in sea level pressure and the

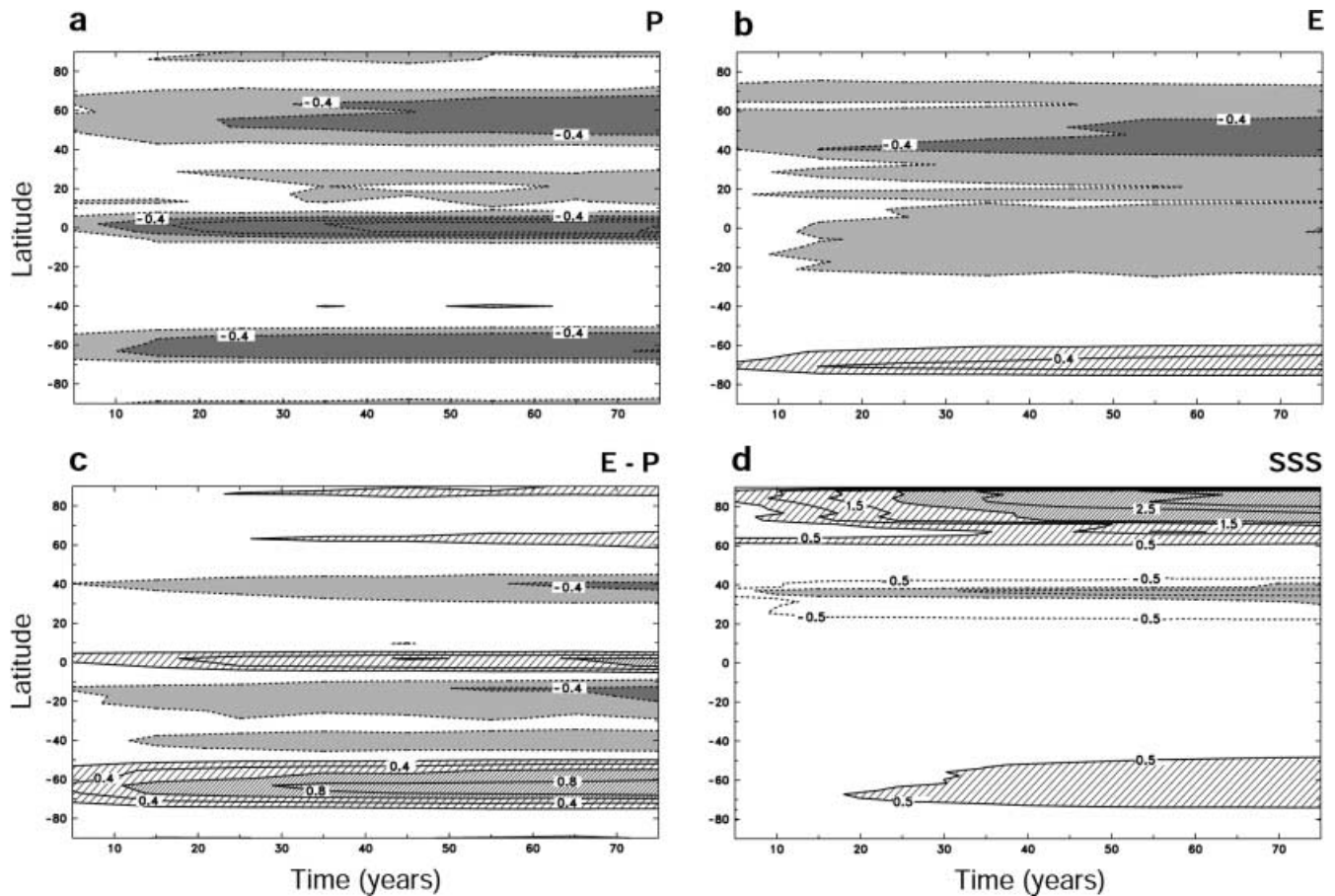


Fig. 14a–d. Temporal evolution of zonally averaged decadal-mean difference between CTR and TRN of, **a** precipitation, **b** evaporation, **c** evaporation minus precipitation, **d** sea surface salinity. Units mm d^{-1} and psu

resulting decreased meridional pressure gradient result in a slackening of the westerlies by as much as 40% over the Antarctic Circumpolar Current (ACC) region (40° to 60°S). The change in wind stress is primarily due to this change in the magnitude of the wind. The southward shift of the maximum peak of zonal wind stress on the northern mid-latitudes is due to the topography of the ice sheet.

4.3.2 Baroclinic and barotropic ocean circulation

The change in the surface wind stress affects on the surface ocean circulation. Figure 18 displays the surface ocean currents in the control simulation and the difference under LGM forcing. The major ocean circulation features such as equatorial currents, the Gulf Stream, the Kuroshio Current, and the ACC are

Fig. 15. Zonally averaged annual-mean salinity difference between CTR and TRN

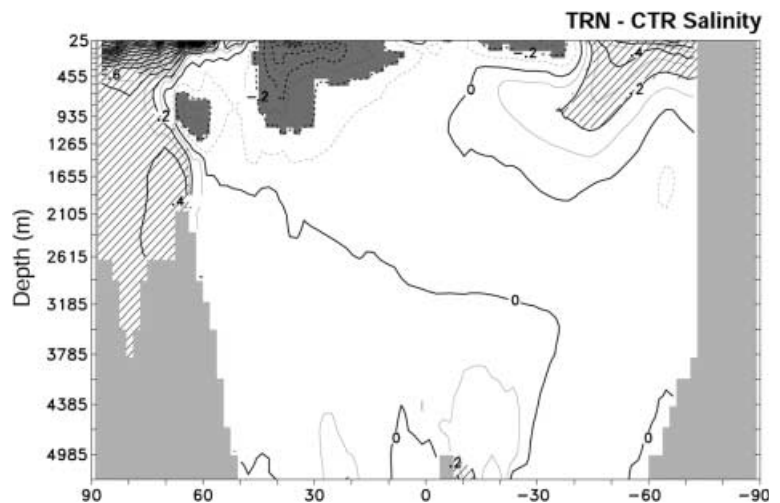
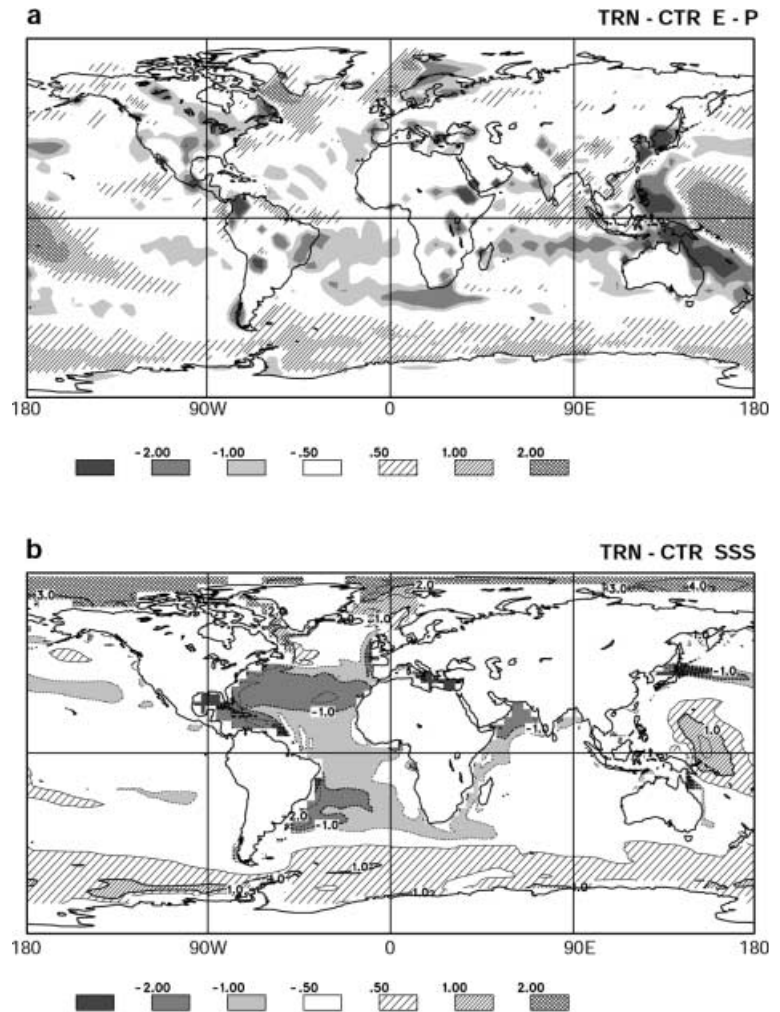


Fig. 16a,b. Horizontal distribution of annual-mean difference between CTR and TRN of, **a** evaporation minus precipitation, **b** sea surface salinity. Units mm day^{-1} and psu



broadly captured in CGCM2 (Fig. 18a). The glacial conditions lead to a substantial intensification of the equatorial current in the Pacific Ocean resulting in the La Niña-like feature noted, and the Gulf Stream and the Kuroshio currents are slightly increased. Due to decreased wind stress over the Southern Ocean, the ACC at the surface decreases somewhat except in the Indian Ocean sector.

The vertically integrated barotropic stream function for CTR and TRN is plotted in Fig. 19. The strength of subtropical gyres is slightly intensified in CGCM2 compared to CGCM1 with simulated Gulf Stream and Kuroshio Current transport of 30 and 40 Sv ($10^6 \text{ m}^3 \text{ s}^{-1}$), respectively. The Indonesian Throughflow, which is a significant component in oceanic meridional heat transport as part of ‘warm-water route’ (Gordon 1986), is about 15 Sv in both CGCM1 and CGCM2. This value is within the uncertainty range of observational estimates (e.g., 10–20 Sv from Fieux et al. 1994; Lukas et al. 1996). The ACC transport produced by CGCM2 is about 80 Sv at the Drake Passage. Although this represents an increase of about 10 Sv relative to CGCM1, it is still too weak compared to the observed estimates

(e.g., 120–140 Sv from Whitworth and Peterson 1985; Read and Pollard 1993; Macdonald and Wunsch 1996).

In TRN, the barotropic transport in the subtropical gyres associated with the Gulf Stream and Kuroshio Current increases to 68 Sv and 47 Sv, respectively. The former is more than double the CTR value while the latter increases by about 17%. The subtropical gyres in the southern Indian Ocean and South Pacific increase by about 40% and 100%. The largest change in barotropic streamfunction is found in the ACC; in the Drake Passage the ACC is more than 140 Sv in TRN, which is a 60 Sv (75%) increase over CTR. The northern North Atlantic subpolar gyre also increases to more than 200% compared to CTR. The Indonesian Throughflow increases by 80% in TRN.

The transient response of the barotropic ACC and the North Atlantic subpolar gyre is clearly not driven by changes in wind stress which decreases over the ACC region and in the latitudes of the subpolar gyres at around 60°N as shown in Fig. 17. This result is somewhat counterintuitive to classical wind-driven ACC theories. Based on the analysis of changes in density structure in the SO (not shown), this appears to be a

Table 1. Amount of river runoff from major drainage basins into the Atlantic and Arctic in CTR and TRN. Units mm day^{-1}

Atlantic	CTR	TRN	Arctic	CTR	TRN
Mississippi	7.1	29.1	Ob	42.5	11.4
Amazon	32.9	63.2	Lena	37.4	28.3
Orinoco	3.0	12.9	Yenisey	25.8	11.4
Parana	15.1	18.8	Mackenzie	13.5	4.2
Congo	14.2	13.4			
Total	72.3	137.4	Total	119.2	55.3

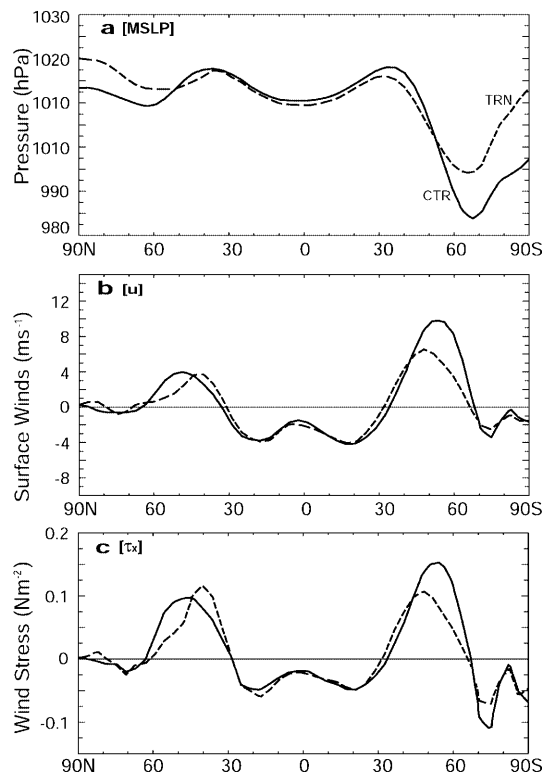


Fig. 17a–c. Latitudinal profile of zonally averaged annual-mean, **a** mean sea level pressure, **b** surface wind, **c** wind stress simulated in CTR (solid) and TRN (dashed). Units hPa , m s^{-1} , and N m^{-2} , respectively

consequence of the enhanced bottom pressure torque due to remarkably increased Antarctic Bottom Water (AABW) density and its formation rate via the joint effect of baroclinicity and relief (JEBAR).

4.3.3 Meridional overturning circulation

Figure 20 presents the global and Atlantic meridional overturning stream functions simulated in CTR and TRN. In CGCM2, the simulated North Atlantic overturning is about 12 Sv (Fig. 20c), which is slightly weaker than observationally based estimates (~ 13 Sv from Dickson and Brown 1994; Schmitz and McCartney 1993; ~ 15 Ganachaud and Wunsch 2000). The observed NADW outflow tends to increase slightly toward the south by entrainment of underlying water mass. For example, at 30°S it is estimated to be about 17–20 Sv

(Rintoul 1991; Zangenber and Siedler 1998; Ganachaud and Wunsch 2000). The simulated NADW layer, however, is too shallow in the North Atlantic and its outflow (~ 10 Sv at 30°S) is underestimated (Fig. 20c).

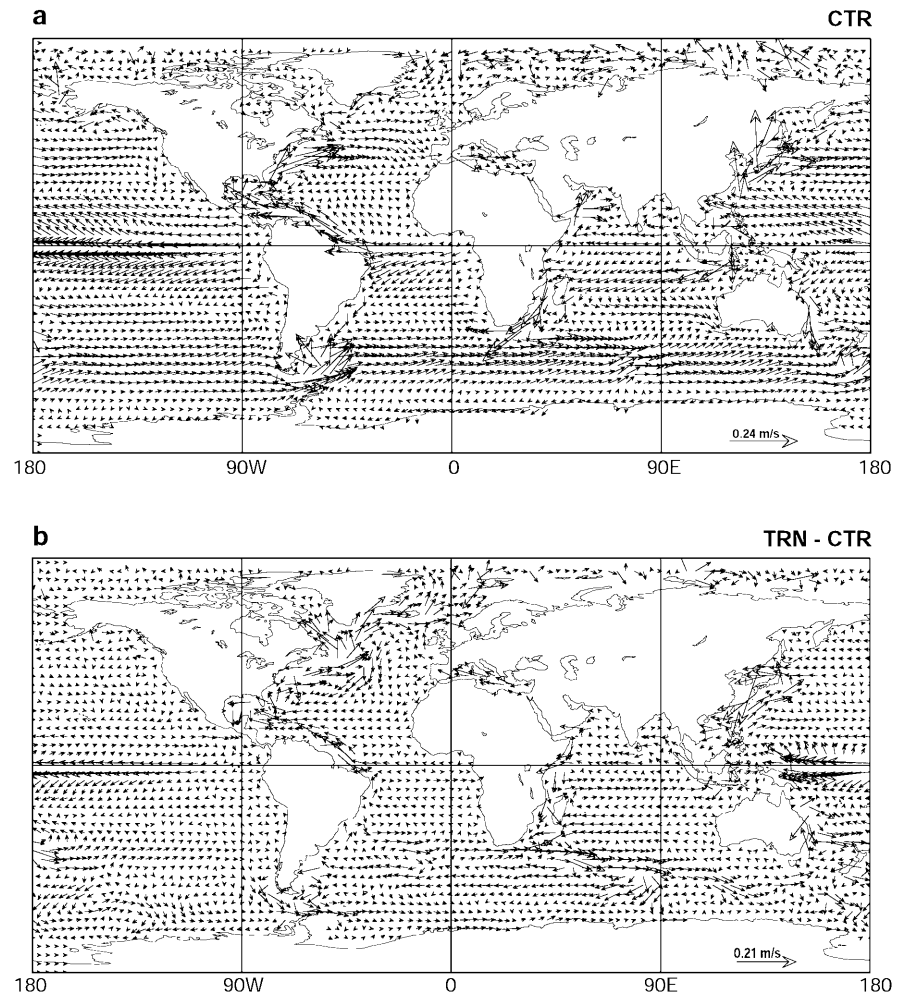
In the Southern Ocean, the simulated overturning circulation associated with AABW formation is about 2 Sv and limited to shallow depths less than 1 km at 70°S (Fig. 20a). This is somewhat weaker than in CGCM1 (~ 6 Sv) and weaker than observed estimates (e.g., 5–10 Sv from Carmack 1977; ~ 8 Sv from Orsi et al. 1999). Note that the uncertainty range of the AABW formation rate is relatively high because most estimates are indirect using chlorofluorocarbon (CFC) uptake or water mass budgets. The simulated AABW outflow (4 Sv) across the equator in the Atlantic basin is, however, very close to observational estimates (McCartney and Curry 1993; Hogg et al. 1999).

The strong positive cell at about 50°S shown in Fig. 20a, referred to as the Deacon cell, is driven by northward Ekman transport in response to SH westerlies (Bryan 1991). It is countered by the eddy induced ‘bolus’ velocity in the GM scheme, which is not included in the stream function shown in this figure. The isopycnal/GM mixing parameterization used in CGCM2 leads to considerable improvement, relative to CGCM1, in deep-ocean water-mass properties. However, the strength of oceanic overturning circulation remains slightly underestimated in comparison to observed estimates.

The overturning circulation becomes much more vigorous in both hemispheres under glacial conditions (Figs. 2f and 20b). This is a result of the intensified oceanic deep convection initiated by the increase of surface ocean density as illustrated Sect. 4.2. In the Atlantic basin, the NADW outflow cell is much deeper (more than 4500 m) and the strength of AABW outflow cell becomes weaker relative to the NADW cell (Fig. 20d). This is due to the increased NADW density relative to that of AABW. In response to the weaker zonal wind stress over the ACC regime, the strength of Deacon cell is remarkably reduced and much shallower in TRN than CTR (Fig. 20a, b).

Combining the oceanic baroclinic and barotropic circulation with meridional overturning circulation, a climatically important feature in terms of the returning path of NADW is captured. From observations, two routes have been suggested to close the NADW outflow. First, Gordon (1986) suggested that the NADW outflow in the Atlantic basin is primarily closed by a ‘warm

Fig. 18. **a, b** Horizontal distribution of annual-mean ocean surface current vectors simulated in CTR. **b** Difference between CTR and TRN. Units m s^{-1}



water route': NADW upwells in the ACC, enters the Indian Ocean from the Pacific through Indonesian Passage, travels around Africa and eventually enters the South Atlantic Ocean. He further suggested that the change in the formation of NADW therefore influences the strength of warm water route. Second, Rintoul (1991) argued that the export of NADW is primarily closed by a 'cold water route': NADW leaving the Atlantic returns as intermediate water entering the Atlantic basin from the Pacific through the Drake Passage. In most OGCMs, the warm water route is not well simulated because of a poor representation of the Agulhas leakage and retroflexion. The NADW export in most OGCMs is indeed closed by the cold water route.

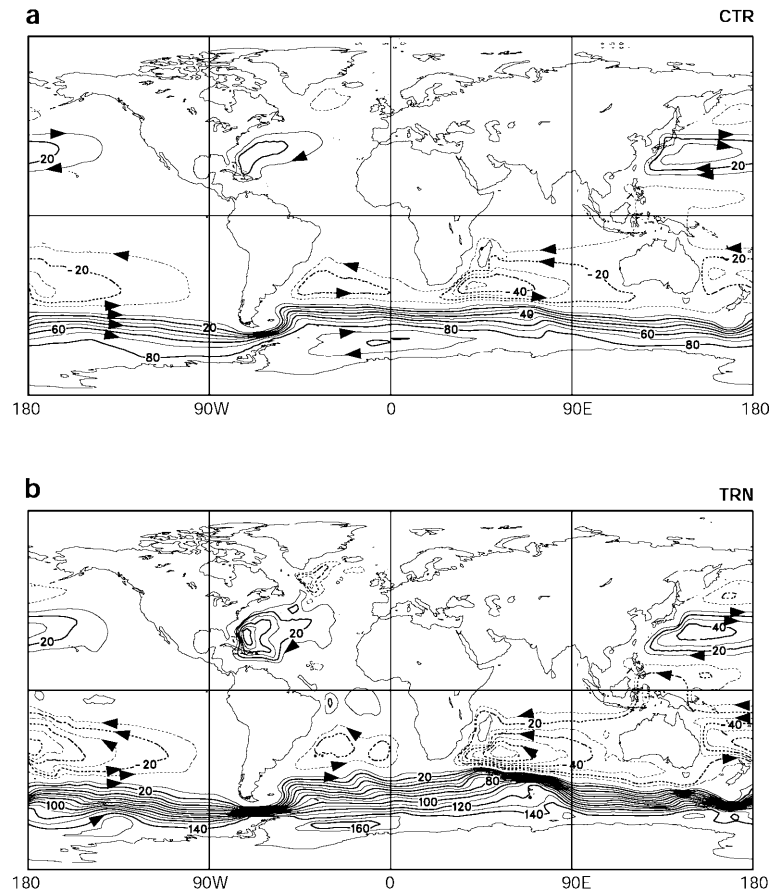
The warm water route is well represented in CTR, however, with warm Pacific water entering the South Atlantic basin via the Agulhas leakage (see Fig. 21a). In TRN, even though the Indonesian Throughflow increases (see also Fig. 19), the southward flow in the Mozambique Channel is blocked at the southern tip of Africa and can not enter the Atlantic and instead it re-joins the eastward ACC (Fig. 21b). The model representation of the Agulhas retroflexion seems to strongly depend on the state of the ACC which is

much stronger and broader in TRN limiting the flow entering the Atlantic. This result is supported by Cai (1994) who was able to reproduce the warm water route when he shut off the ACC. The transient circulation response further confirms Gordon's (1986) prediction, i.e., increased NDAW increases the Indonesian Throughflow as a part of warm water route.

4.4 Meridional heat transport

The oceanic heat transport is a critical component of the global climate system. Although observationally based estimates of heat transport are rather uncertain, the simulated global zonal-mean meridional heat transport of CTR appears to agree reasonably well with observed estimates and the general pattern is close to that inferred from ECMWF re-analysis (Garnier et al. 2000). For example, the maximum northward heat transport is slightly over 1.5 PW ($\text{PW} = 10^{15} \text{ W}$) at around 15°N (Fig. 22). This value is well within the observed error ranges (see Ganachaud and Wunsch 2000, and references therein). In the SH, the maximum southward heat transport appears to be about 1.15 PW at about 15°S ,

Fig. 19a,b. Annual-mean barotropic stream function simulated in, **a** CTR, **b** TRN. Units Sv ($10^6 \text{ m}^3 \text{ s}^{-1}$)



which also agrees reasonably well with observational estimates.

As a result of changes in thermohaline properties and circulation in response to LGM conditions, oceanic meridional heat transport is remarkably altered. In the TRN run, the global meridional heat transport increases in both NH and SH. In the NH, the increase between 30° and 60°N is primarily due to changes in the Atlantic Ocean (Fig. 2b) where northward heat transport increases everywhere due to the increase in North Atlantic overturning discussed previously. On the other hand, the increased southward heat transport in the SH is predominantly due to the change in the Indian Ocean in response to increased Indonesian Throughflow.

5 Comparison to other coupled model results

In this section, we compare the CGCM2 transient climate to the few coupled model LGM results that are available. The differing integration times of the various simulations, which vary from 15 to 700 years, do not present a direct comparison. Nevertheless, it is useful to study the common features of the simulations as well as their differences in response to LGM conditions. Except for the UKMO case, the results represent a transient response because the deep and high-latitude surface ocean requires at least about millennium to reach quasi-

equilibrium. The rapid response of the system to LGM conditions may, however, be compared among the models.

The global and tropical temperature change in response to LGM conditions that is simulated in CGCM2 is similar in magnitude to that of the GFDL coupled model simulation by Bush and Philander (1998) (Table 2). The spatial distribution of SAT and SST change agrees overall in the two cases. For example, the GFDL model exhibits the anomalous warming in the Southern Ocean, even though Bush and Philander (1998) do not discuss it. In both models a patch cooling by about 6°C is seen in the tropical Pacific due to intensified upwelling by stronger trade winds. They also obtained a similar response of sea-ice extent which increased in the NH and decreased in the SH. They regarded the decreased SH ice extent as a LGM state but, based on our analysis, it is seen to be the transient response to an intensified heat transport from the deep to upper ocean as a result of enhanced localized vertical convective mixing.

The CGCM2 transient temperature response differs from that of UKMO (Hewitt et al. 2001). To expedite oceanic adjustment towards the equilibrium Hewitt et al. (2001) cooled the ocean by relaxing to glacial SST obtained from their slab ocean model simulation for the first 70 years and then they coupled to atmospheric model for a further 700 years of simulation. Their results

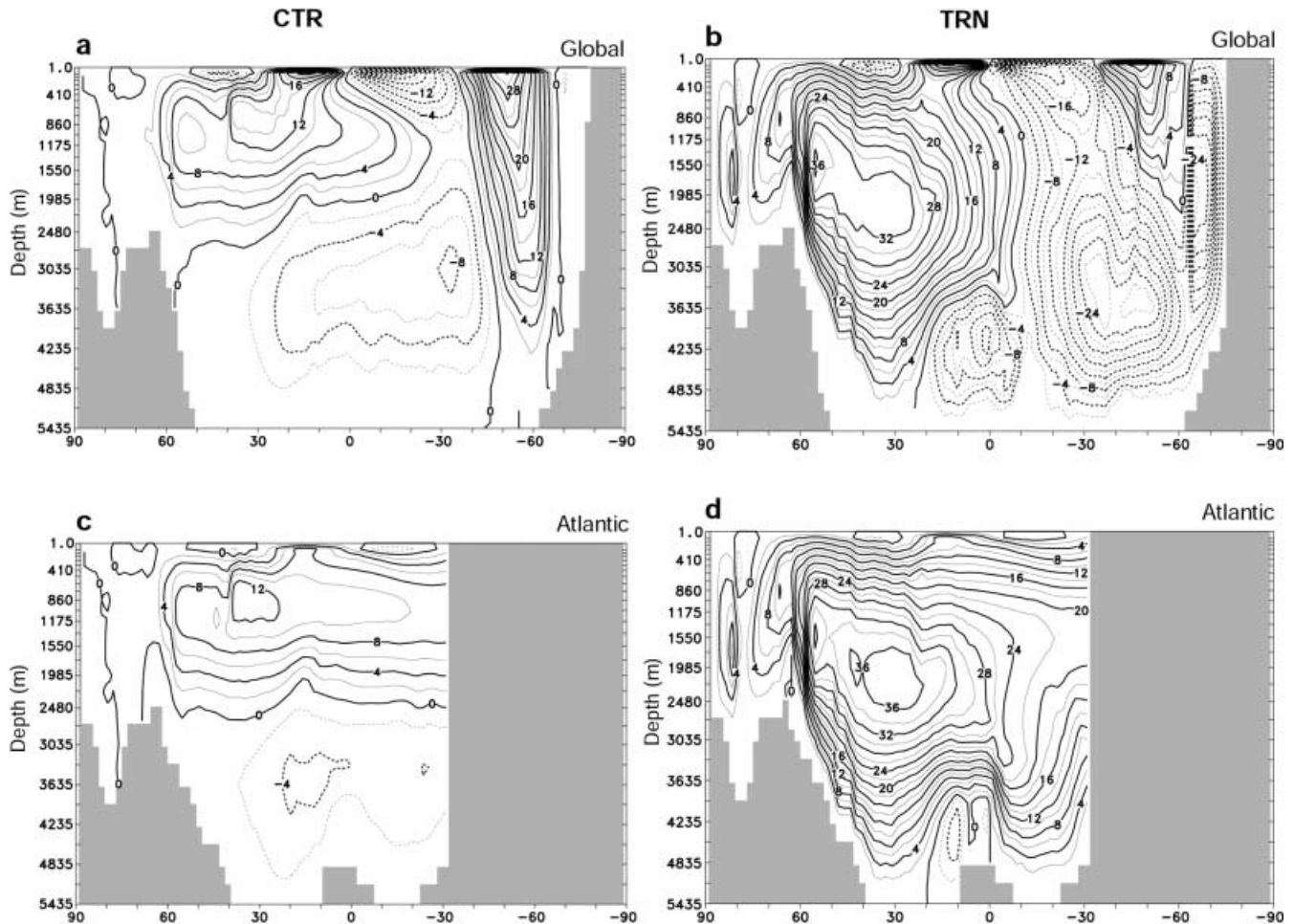


Fig. 20a–d. Annual-mean meridional stream function simulated in CTR for, **a** global, **b** Atlantic; in TRN for, **c** global, **d** Atlantic. Units Sv

are therefore approaching equilibrium. After 770 years, Hewitt et al. (2001) obtained global mean SAT decrease of 3.8 °C, which is smaller than our much shorter time (80 years) temperature response. In their near-equilibrium simulation, SST was slightly warmer in the broader North Atlantic regions than their control run, which they attributed to the increased northward heat transport by increased North Atlantic overturning stream function. Finally, the La Niña-like SST pattern is not obvious in their SAT change although the UKMO model does simulate an El Niño-like response under global warming. Their tropical SAT decrease is overall much smaller than CGCM2.

Using the Japan meteorological research institute (MRI) coupled model, Kitoh et al. (2001) applied additional freshwater correction over the North Atlantic for the first 55 years to obtain more plausible oceanic initial conditions and then integrated for additional 215 years without that freshwater input. After 270 years, they simulated modest cooling in the tropical region as in the case of Hewitt et al. (2001) (Table 2). They obtained anomalous warming patches in the eastern tropical Pacific associated with weaker trade winds. This

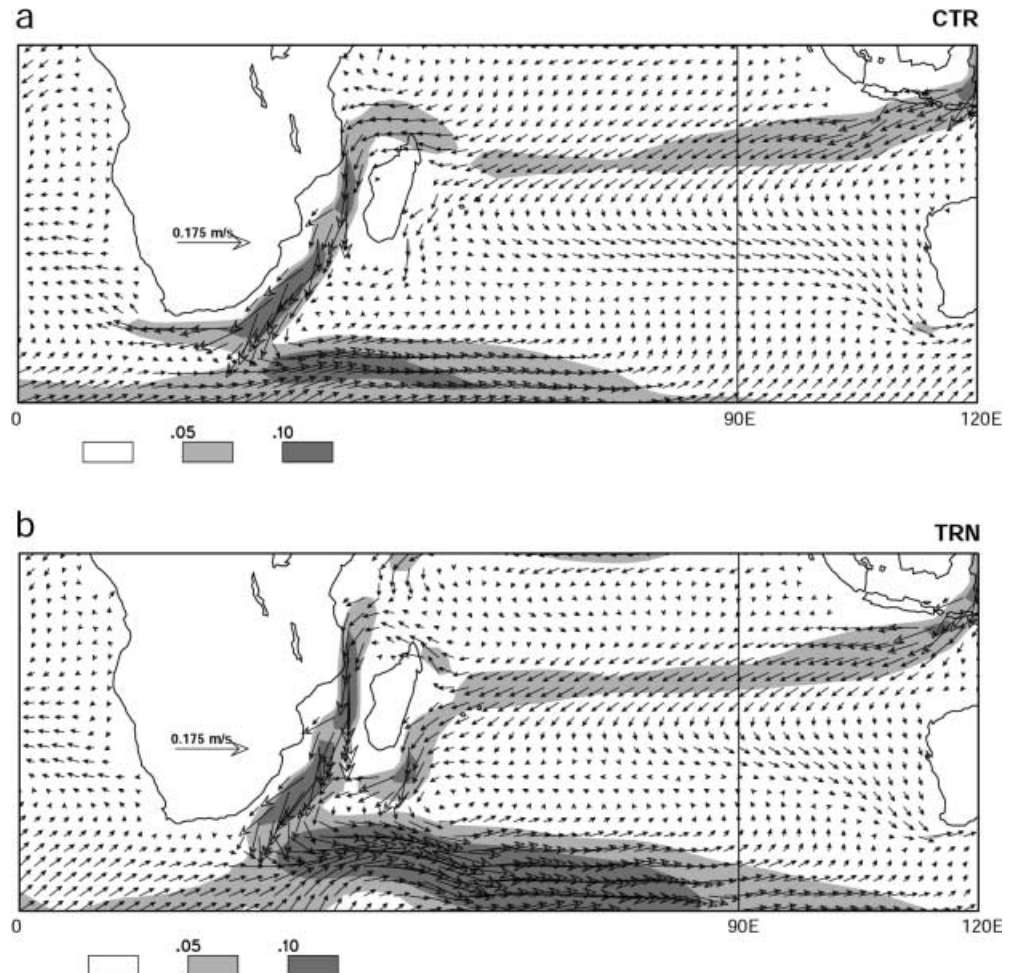
El Niño-like surface temperature response to LGM conditions contrasts with the La Niña-like response in CGCM2, however, is consistent with the La Niña-like response to increased greenhouse forcing of Noda et al. (1999) with opposite polarity.

CLIMAP (1976, 1981) exhibits comparatively weak tropical cooling. The tropical cooling simulated in the UKMO and the MRI coupled models is reasonably close to these CLIMAP estimates. The larger tropical cooling simulated by the CCCma and GFDL coupled models agrees better, however, with the more recent geological proxy evidence (see Sect. 1). These two models also exhibit a strong La Niña-like response in the tropical Pacific which, of course, is absent in the results of mixed-layer ocean LGM simulations, but is indicated especially in the CLIMAP (1976) reconstruction.

6 Summary and conclusions

The evolution of simulated climate toward conditions representative of the Last Glacial Maximum are modeled with the CCCma global climate model (CGCM2).

Fig. 21a,b. Horizontal distribution of annual-mean ocean surface current vectors with magnitude indicated by shading in Indian Ocean sector for, **a** CTR, and **b** TRN



The LGM conditions include a reduction in CO_2 concentration, changes in land boundaries consistent with a 120 m drop in sea level, ice-sheet topography based on reconstructions, and the associated changes in surface albedo and ground cover. The modeled climate initially undergoes a rapid adjustment, followed by a slow evolution toward the LGM equilibrium state. We have examined the climatic changes associated with the initial rapid adjustment 60 to 80 years after the perturbation is imposed. In a companion paper, we analyze the near-equilibrium state.

CGCM2 differs from CGCM1 (Flato et al. 2000; Boer et al. 2000a, b) in that it employs the isopycnal and eddy-induced ocean mixing parametrization of Gent and McWilliams (1990) and the sea-ice dynamics formulation of Flato and Hibler (1992). We provide some comparisons with observations and briefly describe differences with the control climate of the earlier model. The results presented here indicate that CGCM2 generally produces a somewhat more realistic climate than CGCM1, particularly its simulation of ocean temperature and salinity distributions and sea-ice coverage.

Upon imposition of LGM conditions, the modeled climate system evolves rapidly for about 50 years after which, as will be seen in a companion paper, the system

continues to evolve slowly but substantially over the subsequent 1000 years. We examine the short-term evolution in some detail in order to investigate the mechanism governing the transient response of the climate system to the change in forcing. We also compare our results with available results of other coupled model simulations of the transient approach to LGM.

The initial response to the LGM forcing is a rapid decrease in global mean surface air and ocean temperatures of roughly 5 °C and 3 °C. However, this cooling tendency is by no means uniform. In fact, air temperatures over much of the Southern Ocean and portions of the northern North Atlantic actually increase by up to 10 °C. This transient warming is driven by more vigorous oceanic convection that brings heat from the deep ocean to the surface. In the Southern Ocean, this warming leads to an almost complete disappearance of sea ice which further enhances warming via the ice-albedo feedback. Over these regions of anomalous warming the latent and sensible heat flux to the atmosphere increases markedly, resulting in the deep ocean heat loss to the atmosphere and cooling with time.

The hydrological cycle also undergoes a large adjustment with precipitation and evaporation decreasing on average, although evaporation increases over the

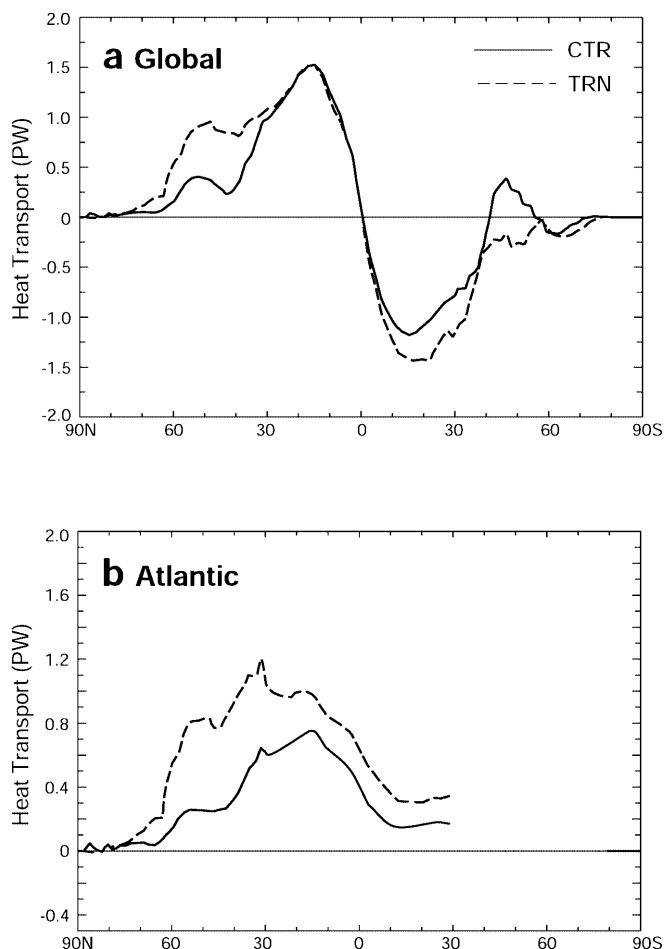


Fig. 22a,b. Zonally integrated annual-mean meridional heat transport simulated in CTR (*solid*) and TRN (*dashed*) for, **a** all ocean basins, **b** the Atlantic. Units PW (10^{15} W)

Table 2. Changes in global-mean surface air temperature (SAT), tropical SAT to LGM boundary conditions simulated in coupled models

Model	Integration time (years)	Global-mean Δ SAT ($^{\circ}$ C)	Tropical Δ SAT ($^{\circ}$ C)
CCCma	80	-4.3	-4.0 ~ -6.0
GFDL	15	-4.3	-5.0 ~ -6.0
UKMO	700	-3.8	-1.0 ~ -2.0
MRI	270	-3.9	+2.0 ~ -2.0

Southern Ocean associated with the anomalous transient warming. The changes in precipitation and evaporation are not uniform however, leading to substantial changes in modeled river runoff. For example, runoff generally increases from river basins draining into the Atlantic, but decreases from river basins draining into the Arctic. Such changes in water budget ultimately affect the sea-surface salinity (SSS), with a decrease in regions of the Atlantic of 1–2 psu, and increases in the Arctic Ocean of more than 3 psu. The decrease in North Atlantic SSS contributes to the gradual weak-

ening of the North Atlantic overturning stream function with time.

In response to the LGM perturbation, the ocean circulation initially becomes more vigorous. For example, the modeled Gulf Stream increases up to double its initial value and the Kuroshio Current increases by about 17%. The Antarctic circumpolar transport increases by about 75% in the Drake Passage (despite a reduction in zonal wind stress in the Southern Ocean). This is presumably due to the enhanced bottom pressure torque associated with the increased AABW density and its formation rate. The North Atlantic subpolar gyre increases from 10 to 30 Sv, and the Indonesian Throughflow increases by 80%. The North Atlantic and Southern Ocean overturning stream function markedly increases from 12 to 36 Sv and from 2 to 26 Sv, respectively, as a consequence of the enhanced oceanic convection.

The CGCM2 transient climate pattern responding to LGM conditions appears to be similar to that of the GFDL coupled model by Bush and Philander (1999), representing larger cooling by more than 5 $^{\circ}$ C in the tropical Pacific due to enhanced trade winds and anomalous warming in the Southern Ocean. On the other hand, Hewitt et al. (2001) using the HadCM3 (their results are closer to equilibrium than other model results) and Kitoh et al. (2001) using the MRI coupled model simulated small tropical cooling less than 2 $^{\circ}$ C and the latter obtained even warming patches in the eastern tropical Pacific.

In conclusion, in our coupled atmosphere–ocean climate model, the climate response to LGM boundary conditions notably different from that of the associated mixed layer ocean version of the model (Vettoretti et al. 2000). Strong dynamic and thermodynamic oceanic feedbacks in fully coupled models give appreciably different LGM results compared to the mixed layer model results in PMIP (Joussaume and Taylor 2000). The most notable oceanic feedbacks to LGM forcing captured in the coupled model are the features associated with the vigorous ocean convection including the anomalously warm surface in the Southern Ocean and the northern North Atlantic and the strongly increased overturning stream function in both hemispheres. These responses are, however, transient features and gradually weaken over time approaching a more moderate or even reversed state, and do not reflect the ultimate LGM equilibrium state as described in detail in the companion paper. Other notable differences include the level of cooling in the tropical Pacific which is considerably cooler in the coupled as compared to the mixed layer case in association with the development of a mean La Niña-like response.

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